

Summary and Recommendation

Telecom Customer Churn Analysis

This analysis examines the churn behavior of telecom customers, identifying key demographic and service-related factors contributing to churn. The dataset revealed several critical trends that can guide strategic decisions to improve customer retention.

Overall Churn Rate

- **26.54%** of customers have churned from the service.
 - This is a significant portion, indicating a need for proactive retention strategies.
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Demographic Insights

- **Gender:** Churn rates between male and female customers are almost identical, indicating that **gender is not a significant factor** in churn.
 - **Senior Citizens:**
 - Senior citizens show a **much higher churn rate**.
 - Around **41% of senior citizens churned**, compared to only **23% of non-senior citizens**.
 - This suggests age-based segmentation strategies might be useful in targeting retention campaigns.
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Tenure

- Customers with **short tenures (1–2 months)** show the highest churn rates.
 - Those with longer tenures (especially **12 months and beyond**) are **far less likely to churn**.
 - Indicates that **early engagement and onboarding** strategies are essential to prevent early churn.
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Contract Type

- Customers on **month-to-month contracts** have a **43% churn rate**.
 - Those on **one-year contracts** show only **11% churn**, and **two-year contracts** show **3.8% churn**.
 - Suggests that promoting longer-term contracts could drastically reduce churn.
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Service-Related Factors

- Customers without **Online Security, Tech Support, or Online Backup** are more likely to churn.
- For example:
 - Among customers **without Online Security**, the churn rate is about **44%**.
 - Those **with Online Security** have a churn rate of just **12%**.
- Similarly, customers **without Tech Support** show a churn rate of **41%**, while those **with support** churn at only **15%**.
- This clearly indicates that **value-added services significantly reduce churn**.

Internet Service Type

- **Fiber optic users** have the **highest churn rate (42%)**, possibly due to cost or service issues.
 - **DSL and No Internet** customers show significantly **lower churn rates** (~15–20%).
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Streaming Services

- StreamingTV and StreamingMovies services show **mixed impact**:
 - While users with these services show moderate churn, **not having them doesn't significantly reduce churn** either.
 - Indicates these features are not primary drivers but could influence customer satisfaction indirectly.
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Key Takeaways

- Focus on **retaining new users** within the first few months by offering onboarding and early engagement benefits.
- Promote **long-term contracts** through discounts or loyalty perks.
- Invest in promoting **online security, tech support, and backup services**, as these clearly correlate with lower churn.
- Consider **targeted retention strategies** for **senior citizens and fiber optic users**, who show higher churn likelihood.